

ZTA Activity Guide

A 60-minute team design clinic · 4 roles · 5 use cases (3 required + 2 bonus). Print this guide and hand one to each team.

The customer & the ask

The customer: a distributed enterprise — corporate HQ and on-prem data centre, remote branches on SD-WAN, a growing hybrid workforce (some running AI agents), and applications across on-prem, AWS and Azure. Grown by acquisition, mid-flight into public cloud; security controls haven't kept pace.

The ask: bring Zero Trust to the environment that already exists — **segment it, see it, and keep enforcing policy** wherever the user, device or workload is. Design a SASE environment satisfying the use cases using ZT Access and Agentic Identity controls (SD-WAN, Cisco ISE, policy, telemetry, and any tool in the Cisco toolbox).

The environment

Network at a glance

- **Campus HQ** — access/distribution/core switching, wired + wireless, large IoT footprint.
- **Branches** — wired + wireless clients over SD-WAN.
- **Data center** — on-prem apps plus workloads in public cloud.
- **Wireless** — corporate SSIDs and a Guest SSID.
- **Identity & policy** — the customer can run Cisco ISE for auth, profiling and posture.

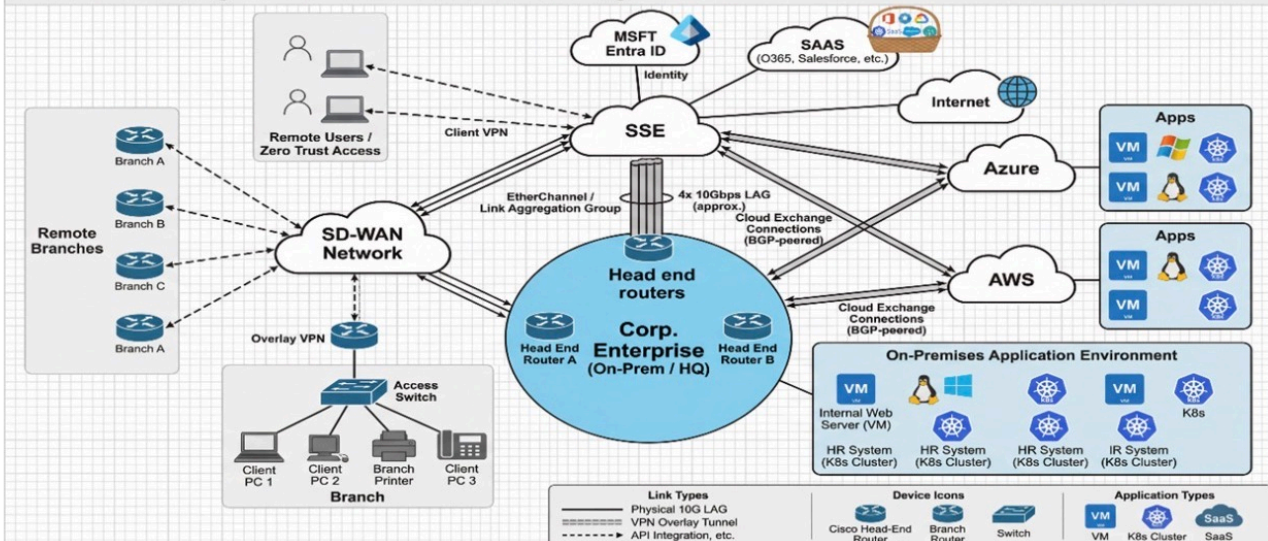
Who's on the network

- **Employees & trusted devices** — laptops, mobiles (Group=Employees).
- **IoT** — printers, cameras, badge readers (Group=IOT).
- **Deskside agents** — dedicated Mac-minis (Group=deskagents).
- **Guests** — Internet-only (Group=Guest).
- **Corporate PCs** — some behind on AntiVirus / OS updates.

Reference architecture

Everyone starts from the same picture. Your diagram for each use case is your version of this, with the products and policy you propose drawn on top.

Architecture Design Clinic - Reference Network Diagram



Business objectives

- **Secure, scalable segmentation** — put every device in the right group and control what it can reach, without re-architecting VLANs/subnets.
- **Pass compliance (PCI)** — demonstrate segmentation and control of the cardholder environment.
- **Cut operational toil** — automate profiling and posture.
- **Work within budget** — reuse the existing network for enforcement.
- **Improve endpoint hygiene** — bring corporate PCs up to date before full access.
- **Complete the required cases, then bonus** — Use Cases 1–3 are required, 4–5 are bonus; each scored on the leaderboard.

Four roles, no passengers

OWNS THE PACE

IT Director

Watches the clock and keeps the team moving; calls time checks; confirms each use case is submitted before moving on.

OWNS THE CHALLENGE

Digital Resiliency Officer

Challenges each decision (“what would make this fail?”) to spark dialogue, surfaces the threat surface, makes the team justify trade-offs — not to block.

OWNS THE DESIGN

Network Architect

Holds the pen and draws the topology; ensures every product/control has a home on the diagram; decides placement and trust boundaries.

OWNS THE WRITE-UP

Network Security Engineer

Captures decisions; owns the pain → product mapping and product write-up; runs the diagram upload and final submission.

How the hour runs

- **Teams of 4** — one person per role. Your 60-minute clock starts once all four seats are filled.
- **Five use cases, in order** — 1–3 required, 4–5 bonus; solve Use Case 1 first, submitting one unlocks the next.
- **Required first, then bonus** — complete the three required cases, then bank bonus rounds for extra points.
- **For each use case** — map pain → product, list products (how & why), draw a topology diagram, then submit as a team.

Segmentation methodology

Shared vocabulary — reference, not the answer. Apply it to each use case.

Why VLANs / subnets alone struggle

- Segmentation tied to IP/VLAN means topology changes ripple everywhere.
- Access rules become long IP ACLs that are hard to read, audit and maintain.
- Every new device type needs new subnets and rules; staff can't keep up.

Identity-based (group) segmentation

- Classify each user/device into a group at the point of access (authentication + profiling).
- Carry the group as a tag with the traffic, independent of IP / VLAN / topology.
- Write policy as group-to-group, enforced across the existing network.

The building blocks

- Authenticate & profile every endpoint (802.1X, MAB, web auth).
- Posture — check endpoint health (AV, OS, patches) and remediate before full access.
- Tag & enforce — assign a group tag; switches / routers / firewalls enforce inline.
- Share context — feed identity/group to firewalls, DC, SD-WAN and threat tools.

How to approach each use case

- 1 · Map pain → product. For every pain point, name the product/feature that fixes it.
- 2 · Say how & why. One line on how it works and why it fits (differentiators welcome).
- 3 · Read the task & parameters. Note the specific groups, access rules and constraints.
- 4 · Propose a diagram. Draw the topology with your solution on it, then submit.

What “good” looks like

Illustrative example on a **different** scenario — calibrate depth and format, don't copy.

Customer pain: “Guests and IoT cameras share the same VLAN, so a compromised camera can reach guest laptops — and we’ve no budget for more firewalls.”

CUSTOMER PAIN POINT	PROPOSED PRODUCT(S)	HOW IT ADDRESSES / WHY
IoT and Guest share a VLAN; a hacked camera can reach guests	Cisco ISE + TrustSec SGTs	Profile each device at onboarding, tag IOT and Guest, enforce an SGACL that denies IOT ↔ Guest on existing switches.
No budget for more firewalls to segment	TrustSec SGACL enforcement	Group-to-group policy enforced inline on the Catalyst switches you already own.

A strong diagram labels every group, marks the enforcement point (where the SGACL/firewall sits), and traces one flow end-to-end (e.g. camera → denied → guest). Legible beats pretty.

The use cases

Use Case 1 — Managed Environment — Segmentation

Required

The customer wants to segment campus & branch wired/wireless client devices into logical groups and control what each group can reach.

Customer requirements

- Segment campus & branch wired and wireless clients into logical groups: Employees, IOT, deskagents, Guest.
- Employees = laptops, mobiles (Group=Employees). IoT = printers, cameras, badge readers (Group=IoT). Deskside = dedicated Mac-minis (Group=deskagents). Guest SSID = Internet only (Group=Guest).
- IoT restricted from the other groups but needs data center and Internet access.
- Guests get Internet access only.
- Employees and IoT require controlled Internet and data center application access.
- Deskside Mac-minis reach the on-prem data center only.

Customer pain points

- Prior segmentation failed — using VLANs/IP subnets is complex and tedious.
- Profiling is labour-intensive; staff unable to keep up with constant new devices joining.
- Failing PCI audits.
- Corporate PCs not running the latest AntiVirus and OS updates.
- No budget for buying more firewalls to do segmentation.

Your tasks

- Draw a diagram of the customer's campus and branch wired/wireless networks.
- Create your proposed segmentation solution; be ready to explain the product selected, how it works, and why (include Cisco differentiators if available).
- Make sure your proposed solution addresses each customer requirement and pain point.
- Attach the diagram and submit for SME review and scoring.

Use Case 2 — Use Case 2

Required

This use case is a placeholder for the demo — requirements, pain points and tasks will be added before it goes live.

Use Case 3 — Use Case 3

Required

This use case is a placeholder for the demo — requirements, pain points and tasks will be added before it goes live.

Use Case 4 — Use Case 4

Bonus

This use case is a placeholder for the demo — requirements, pain points and tasks will be added before it goes live.

Use Case 5 — Use Case 5

Bonus

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